

CRT SECOND DAY

1. write a cpp program that takes a person's age as a input.

if the age is 18 or older. print eligible to vote

```
Sol: #include <iostream>

using namespace std;

int main()
{
    int a;
    cout<<"Enter your age: "<<endl;
    cin>>a;
    if (a >=18)
    {
        cout<< "Eligible to vote"<<endl;
    }
    else
    {
        cout<<"Not eligible to vote"<<endl;
    }
    return 0;
}
```

2. write a cpp program that takes an integer input representing a persons age. use an else if ladder to determine and print. eligible to vote if age is 100 nd more in century category. eligible to vote = 18-99. not eligible to vote if it is between 0 and 17. invalid if age is negative

```
#include <iostream>

using namespace std;

int main()
{
```

```

int a;

cout<<"Enter your age: "<<endl;

cin>>a;

if (a >=100)
{
    cout<< "Eligible to vote- Century category"<<endl;
}
else if( a >=18 && a<=99)
{
    cout<<"eligible to vote"<<endl;
}
else if (a>=0 && a<=17)
{
    cout<<"Not eligibel to vote"<<endl;
}
else
{
    cout<<"your age is negative"<<endl;
}

return 0;
}

```

Q3. write a cpp program that takes 3 distinct int as input and find the largest among them using nested conditional statements

```

#include <iostream>

using namespace std;

```

```
int main()
{
    int a, b, c;
    cout<<"Enter the distinct integer value: "<<endl;
    cin>>a>>b>>c;
    if (a > b)
    {
        if (a > c)
        {
            cout<<"The greater value is" << a <<endl;
        }
        else
        {
            cout<<" The greater value is: "<< c <<endl;
        }
    }
    else{
        if (b>c)
        {
            cout<<"The greater value is "<<b<<endl;
        }
        else {
            cout<<"the greater value is "<<c<<endl;
        }
    }
    return 0;
}
```

ques 4. write a cpp program that takes 2 int as a input from user, find max of 2 using ternary conditional operators.

```
#include <iostream>

using namespace std;

int main()
{
    int a, b;
    int result;
    cout<<"Enter the distinct integer value: "<<endl;
    cin>>a>>b;
    (a>b) ? a : b;
    result = (a>b) ? a : b;
    cout<<"The greatest digit is: "<< result<<endl;

    return 0;
}
```

ques 5. write a cpp program that takes an integer a as an input.

print all number from 1 to n but skip any number that is multiple of 3 using continued statements.

Sol: #include <iostream>

using namespace std;

```
int main()
{
    int n;
    cout<<"Enter the integer value: "<<endl;
    cin>>n;
```

```

for (int i = 1; i<=n; i++)
{
    if ( i %3==0)
    {
        continue;
    }
    else{
        cout<<"the number: "<< i <<endl;
    }
}
return 0;
}

```

ques 6. write a cpp program that continusly takes a positive integer as input from the user and add them to running a sum variable if the users enters negative number, immediately terminate the loop using break statement and print the final sum.

SOL: #include <iostream>

using namespace std;

```

int main()
{
    int n;
    int sum;
    sum = 0;

    while(true)
    {
        cout<<"Enter integer: "<<endl;
        cin>>n;
        if ( n<0)

```

```

{
    break;
}
else {
    sum= n ;
}
cout<<"final sum is" <<sum<<endl;
}
return 0;
}

```

ques 7. WAP that takes a student percentage between 0 to 100 as input and calculates their grades in the following crireria.

A- above 90

B-80-89

C-70-79

D-60-69

below 60- fail

SOL: #include <iostream>

using namespace std;

int main()

{

int n;

cout<<"Enter integer: "<<endl;

cin>>n;

if (n > 100 || n<0)

{

```
    cout<< "Invalid number"<<endl;
}
else if (n>90)
{
    cout<<"A"<<endl;
}
else if (n>=80 && n<=89)
{
    cout<<"B"<<endl;
}
else if (n>=79 && n<=79)
{
    cout<<"C"<<endl;
}
else if (n>=60 && n<=69)
{
    cout<<"D"<<endl;
}
else
{
    cout<<"fail"<<endl;
}
return 0;
}
```

ques 8. WAP that takes a year as a input and check if it is a leap year or not.
it should be divisible by 4 but not be divisible by 100 except a century year
divisible by 400

```
SOL: #include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int n;
```

```
    cout<<"Enter year: "<<endl;
```

```
    cin>>n;
```

```
    if (( n%4==0 && n%100 != 0) || (n%400==0))
```

```
    {
```

```
        cout<< "It is a leap year"<<endl;
```

```
    }
```

```
    else
```

```
    {
```

```
        cout<<"It is not a leap year"<<endl;
```

```
    }
```

```
    return 0;
```

```
}
```

ques 9. WAP that takes a single alphabet char as a input from the user and determines whether it is vowel and consonants

```
Sol: #include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    char ch;
```

```
    cout<<"Enter character: "<<endl;
```



```
cin>>ch;
if(! ((ch>= 'a' && ch<= 'z') || (ch>= 'A' && ch<= 'Z'))
{
    cout<<"Invalid output"<<endl;
    return 0;
}
```

```
switch(ch)
{
    case 'A':
        cout<<"Vowel"<<endl;
        break;
    case 'E':
        cout<<"Vowel"<<endl;
        break;
    case 'I':
        cout<<"Vowel"<<endl;
        break;
    case 'O':
        cout<<"Vowel"<<endl;
        break;
    case 'U':
        cout<<"Vowel"<<endl;
        break;
    case 'a':
        cout<<"Vowel"<<endl;
        break;
    case 'e':
        cout<<"Vowel"<<endl;
```

```

break;

case 'i':

cout<<"Vowel"<<endl;

break;

case 'o':

cout<<"Vowel"<<endl;

break;

case 'u':

cout<<"Vowel"<<endl;

break;

default:

cout<<"It is a consonant"<<endl;

}

return 0;

}

```

ques 10. write a cpp program that stimulates a simple customer support system using switch case statement in that menu options are one for support

2 for billing

3 for technical issue

0/4 to speak to an agent

Sol: #include <iostream>

using namespace std;

int main()

{

int a;

cout<<"Enter options: "<<endl;

```

cin>>a;
switch(a)
{
    case 2:
        cout<<"Billing"<<endl;
        break;
    case 3:
        cout<<"Technical issue"<<endl;
        break;
    case 0:
        cout<<"speak to an agent"<<endl;
        break;
    case 4:
        cout<<"Speak to an agent"<<endl;
        break;

    default:
        cout<<"number is not valid"<<endl;
}
return 0;
}

```

ques 11. WAP that stimulates a ATM withdrawl system, the program should take the withdrawl amount as a input from user and determines whether the transaction is valid or not.

- amount should be a multiple of 100(ATM RULE)
- should be +ve
- less than and equal to balance

Sol: #include <iostream>

using namespace std;

```
int main()
{
    int balance= 10000;
    int amount;

    cout<<"Enter withdrawl amount: "<<endl;
    cin>>amount;

    if (amount<= 0)
    {
        cout<<"invalid amount"<<endl;
    }
    else if (amount % 100 !=0)
    {
        cout<<"Cant withdrawl it should be a multiple of 100"<<endl;
    }
    else if( amount> balance)
    {
        cout<<"insufficient balance"<<endl;
    }
    else
    {
        balance = balance-amount;
        cout<<"Transaction successful"<<endl;
        cout<<"Your remaining balance is: "<< balance<< endl;
    }
    return 0;
}
```

